

EE5203 L01 Embedded C Readiness Diagnostic

Where: CATS, Tests & Quizzes, week 1 (after L01, before L02)

Time: 15 minutes, one attempt

Grading: Not graded. Your answers only choose which C practice track you get.

Some questions are about topics we have not taught yet. That is expected. Answer what you can and leave the rest blank. A blank answer is useful information.

Part A - Trace familiar programming ideas

1. What is the final value of `x`?

```
int x = 3;
x = x + 2;
x = x * 2;
```

2. What is the final value of `status`?

```
int temperature = 18;
int status = 0;

if (temperature > 20) {
    status = 1;
}
```

3. List the values printed or observed during this loop.

```
for (int i = 0; i < 4; i++) {
    /* observe i */
}
```

4. Complete the function so it returns the larger input.

```
int maximum(int a, int b)
{
    /* your code */
}
```

5. The array contains four samples:

```
int samples[4] = {10, 20, 30, 40};
```

- a. What is `samples[0]`?
- b. What is `samples[3]`?
- c. Which index would be invalid?

Part B - Starting representation knowledge

6. Write decimal 13 in binary.
7. Write binary 1010 in decimal.
8. Which hexadecimal digit represents decimal 15?
9. Without calculating the final numeric value, describe what this expression does to bit 3 of `value`:

```
value | (1U << 3)
```

Choose one: **sets it** / **clears it** / **toggles it** / **not sure**.

Part C - Debugging

10. A compiler reports errors on lines 8, 12, and 20. Which error should normally be investigated first, and why?

11. In one sentence, distinguish a syntax error from a logic error.